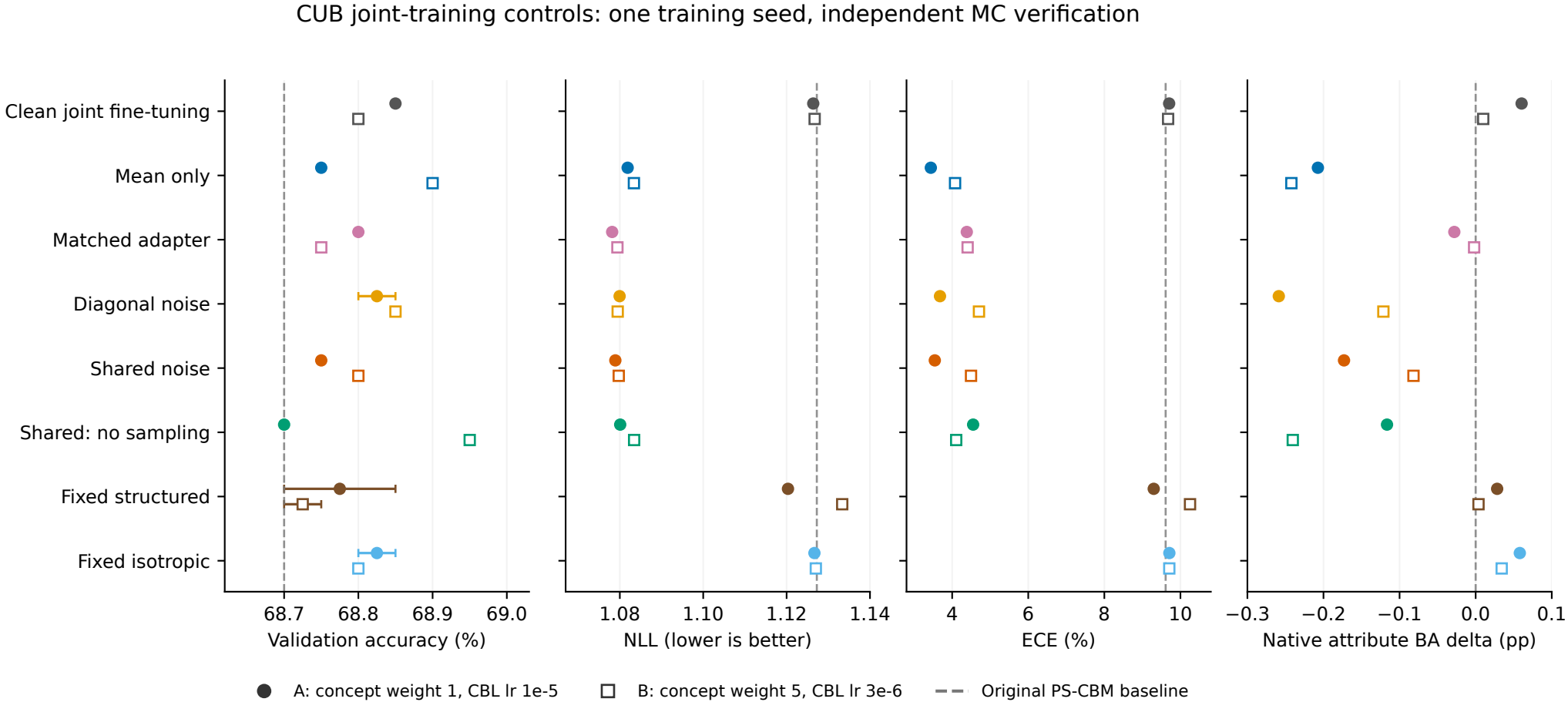
CUB joint-training controls: one training seed, independent MC verification
Clean joint fine-tuning
Mean only
Matched adapter
Diagonal noise
Shared noise
Shared: no sampling
Fixed structured
Fixed isotropic
68.7
68.8
68.9
69.0
Validation accuracy (%)
1.08
1.10
1.12
1.14
NLL (lower is better)
4
6
8
10
ECE (%)
−0.3
−0.2
−0.1
0.0
0.1
Native attribute BA delta (pp)
A: concept weight 1, CBL lr 1e-5
B: concept weight 5, CBL lr 3e-6
Original PS-CBM baseline
Whiskers: two independent 256-sample MC runs, not confidence intervals. Native BA: train-only mapping and thresholds, 68/312 attributes.